Modeling Linear Sliding Electrostatic Motors With Skewing

Sohee John Yoon, Inrak Choi, and Yong-Lae Park

***Abstract*—Linear sliding electrostatic motors operate based on the Coulomb force between charges in repeatedly patterned electrodes in a slider and stator configuration. We present a model for linear electrostatic motors derived from the assumptions of the electric fields that formally provide the physical relationship between the motor design and the capacitance matrix. The cross-sectional actuation force is obtained with the capacitance matrix and explored for various motor parameters. We also extend the model for generic skewing patterns, five of which are shown representatively with their influences on the actuation force. The experimental results validate the derived capacitance matrix, the model for skewing patterns, and the resultant model of the actuation force. Overall, our model provides an insight into the effects of the parameters involved in the derivation of the actuation force.**

***Index Terms*—Electrostatic actuator, experimental validation, modeling, skewing.**

I. Introduction

ELECTROSTATIC actuators operate by the electrostatic force (i.e., Coulomb force) between two charged particles. The simplicity of this working principle has led to their attractive uses in MEMS devices in the microscale [1], [2], [3], [4], [5] as well as in larger devices [6], [7], motors [8], [9], [10], robots [11], [12], [13], [14], [15], and wearables [16], [17], [18]. Among these, linear sliding electrostatic motors have been designed to provide actuation force over an indefinite stroke by using repeatedly patterned electrodes with alternating voltages in a slider and stator configuration [19], [20], [21], [22], [23].

Manuscript received 15 January 2024; revised 19 May 2024 and 13 June 2024; accepted 7 July 2024. Date of publication 6 August 2024; date of current version 24 December 2024. This work was supported in part by the National Research Foundation under Grant 2021R1A2C2093790 and Grant RS-2023-00208052 and in part by the Technology Innovation Program under Grant 20008912 both funded by the Korean Government, MSIT & MOTIE, respectively. *(Sohee John Yoon and Inrak Choi contributed equally to this work.) (Corresponding author: Yong-Lae Park.)*

Sohee John Yoon and Yong-Lae Park are with the Department of Mechanical Engineering, Institute of Advanced Machines and Design (IAMD) and the Institute of Engineering Research, Seoul National University, Seoul 08826, Republic of Korea (e-mail: jylight@gmail.com; ylpark@snu.ac.kr).

Inrak Choi is with Intuitive Surgical, Inc., Sunnyvale, CA 94087 USA (e-mail: inrak.choi@intusurg.com).

Digital Object Identifier 10.1109/TIE.2024.3429577

Models for such motors in the literature exist [24], [25], [26], [27], [28]; but although they are sufficient to provide a standard with which experimental results may be compared, they are based on approximations [26], curve fittings of experimental measurements [27] or numerical simulations [28] and do not provide a physical approach based on the assumptions of the electrical fields. Wang et al. used a method of moments and Coulomb's law to provide a physics-based model [28] but did not characterize the form of the capacitance matrix. Additionally, the effects of skewing the electrodes have been studied [22], [24], [27] but only for linear skewing patterns.

Here, we model the electromechanical force of linear electrostatic motors based on the assumptions of the electric fields. Assuming two different shapes for these fields between the electrodes of the stator and that of the slider such that they are 1) projected by neighbors, or 2) strictly vertical, we derive the formal representation of the capacitance matrix based on the field shapes in terms of the motor design parameters. From the capacitance matrix we then obtain the cross-sectional force. Because the capacitance matrices in our model are based solely on assumptions on the shapes of the electrical fields, their derivation is obtained in a straight-forward manner from the design parameters instead of approximations, experimental measurements, or numerical simulations. Consequently, our modeling approach retains the relationship between the design parameters and the actuation force.

Furthermore, we present the modeling of generic skewing patterns beyond linear skewing. Five representative patterns are shown with their influences on the actuation force and the ripple rate. Using the fabricated prototypes, we experimentally validate our actuator force and skewing pattern models for the two electric field assumptions. Within a reasonable level of error, the models correctly predict the actuator force.

II. Theory and Calculation

The cross-section of a generic linear electrostatic motor in a slider and stator configuration with repeatedly patterned electrodes is illustrated in Fig. 1. To prevent electrical shorting between the electrodes, an insulating film is used to encapsulate the electrodes. Additionally, a dielectric material that is either fluid (air or oil as a lubricant) [12], [22], [26] or solid (glass beads to reduce frictional loss) [11], [12], [13], [14], [26], [27] is typically placed between the slider and the stator. The electrodes are assumed to be ideal conductors (electrical

0278-0046 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:13:56 UTC from IEEE Xplore. Restrictions apply.

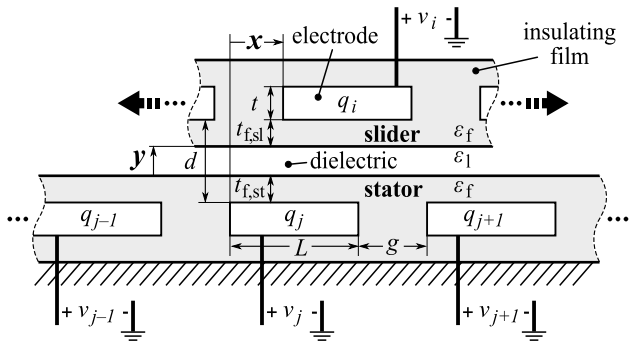


Fig. 1. Cross-section of the linear electrostatic motor with charge q_i in the ith electrode of the slider and the charge q_j in the jth electrode of the stator supplied by voltages v_i and v_j, respectively. Electrodes have length L and thickness t and are equally spaced apart by gaps g such that they repeat every pitch p $(= L + g)$. The insulating film and separating dielectric material have dielectric constants ε_f and ε_l, respectively. Unilateral film thicknesses of the slider $t_{\mathrm{f,sl}}$ and stator $t_{\mathrm{f,st}}$ combined with the separation distance y equal the transversal distance d. The position of the slider with respect to the stator is given by x.

conductivity $\sigma \to \infty$) and the insulators to be ideal insulators ($\sigma = 0$).

In a quasistatic linear electric field system described by $\boldsymbol{q} = \mathbf{C}\boldsymbol{v}$, the actuator force $\boldsymbol{f}$ is derived as

$$\boldsymbol{f} = \int \nabla \boldsymbol{q}\, \mathrm{d}\boldsymbol{v} = \frac{1}{2}\boldsymbol{v}^{\mathrm{T}} \nabla \mathbf{C} \boldsymbol{v} \tag{1}$$

where $\nabla = (\partial/\partial x,\ \partial/\partial y) \in \mathbb{R}^2$ is the 2D gradient operator.

Primarily, linear electrostatic motors are used for the actuation force generated along the direction of sliding, which here corresponds to f_x. The other force f_y is the normal force between the slider and the stator, which although does not directly contribute to the actuation force, may still affect it through the frictional effects or by changing the transversal distance d between the electrodes of the slider and those of the stator ($d = t_{\mathrm{f,sl}} + t_{\mathrm{f,st}} + y$).

While (1) is presented in the literature [26], [27], these models do not derive the capacitance matrix directly from the shapes of the electric fields through the irrotational assumption from Coulomb's law ($\nabla \times \boldsymbol{E} = 0$) and Gauss' law ($\nabla \cdot \boldsymbol{E} = \rho_\mathrm{e}/\varepsilon$). In the following section, we include the derivation of the capacitance matrix and the consequent actuation force based on two different assumptions on the shapes of the electric fields.

A. Capacitance

Calculation of (1) requires the knowledge on the capacitance matrix $\mathbf{C}$ that, combined with the input voltages $\boldsymbol{v}$ consisting of the slider voltages $\boldsymbol{v}_{\mathrm{sl}}$ and the stator voltages $\boldsymbol{v}_{\mathrm{st}}$ such that $\boldsymbol{v} = (\boldsymbol{v}_{\mathrm{sl}}, \boldsymbol{v}_{\mathrm{st}})^{\mathrm{T}}$, provides the force. This capacitance matrix $\mathbf{C}$ for linear systems is symmetric and expressed in the form of

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_\mathrm{s} & -\mathbf{C}_\mathrm{m} \\ -\mathbf{C}_\mathrm{m}{}^{\mathrm{T}} & \mathbf{C}_\mathrm{s} \end{bmatrix} \tag{2}$$

where $\mathbf{C}_\mathrm{s}$ is the self-capacitance matrix that relates the slider-to-slider and the stator-to-stator charges, and $\mathbf{C}_\mathrm{m}$ is the mutual capacitance matrix that relates the slider charges to the stator charges and vice-versa.

The linear electrostatic motors in the literature are commonly fabricated using the flexible printed circuit board (FPCB) technology, resulting in small electrode thicknesses t such that $L \gg t$ may be assumed, allowing the fringing effects to be neglected in the model. The electric fields are then simplified to be uniform and confined within the regions that connect the electrodes between the slider and the stator. These regions are modeled by the following two different approaches.

1) Fields Are Projected by Neighbors (FPN): Electric fields are confined to the regions that are parallelogram projections from m neighboring electrodes between the slider and the stator. Fig. 2 depicts such regions as being bounded by the dashed lines. The corresponding capacitance matrix written as (2) has self and the mutual capacitance matrices $\mathbf{C}_\mathrm{s}$ and $\mathbf{C}_\mathrm{m}$ which are circulant matrices with $\boldsymbol{c}_\mathrm{s}$ and $\boldsymbol{c}_\mathrm{m}$ centered along their diagonals, respectively; therefore, they act as discrete circular convolution operators for $\boldsymbol{c}_\mathrm{s}$ and $\boldsymbol{c}_\mathrm{m}$. Expressly,

$$\boldsymbol{c}_\mathrm{s} = \left(-\frac{\varepsilon_\mathrm{f} t}{g+mp}, \ldots, -\frac{\varepsilon_\mathrm{f} t}{g}, C_\Sigma, -\frac{\varepsilon_\mathrm{f} t}{g}, \ldots, -\frac{\varepsilon_\mathrm{f} t}{g+mp} \right) \tag{3}$$

and

$$\boldsymbol{c}_\mathrm{m} = C_{\mathrm{series}} \left(\frac{d^2}{d^2+(x+mp)^2}, \ldots, \frac{d^2}{d^2+x^2}, \ldots, \frac{d^2}{d^2+(x-mp)^2} \right) \tag{4}$$

where m is the number of charges ("neighbors") to the right and the left of q_i or q_j that are included in the model. C_Σ is the sum of the absolute values of all the other terms in both $\boldsymbol{c}_\mathrm{s}$ and $\boldsymbol{c}_\mathrm{m}$ such that

$$\begin{aligned} C_\Sigma = {} & \frac{\varepsilon_\mathrm{f} t}{g+mp} + \ldots + \frac{\varepsilon_\mathrm{f} t}{g} + \frac{\varepsilon_\mathrm{f} t}{g} + \ldots + \frac{\varepsilon_\mathrm{f} t}{g+mp} \\ & + \frac{d^2}{d^2+(x+mp)^2} + \ldots + \frac{d^2}{d^2+x^2} + \ldots \\ & + \frac{d^2}{d^2+(x-mp)^2} \end{aligned} \tag{5}$$

and C_{series} is the equivalent series capacitance between the slider and the stator electrodes along the transversal distance d such that

$$\frac{1}{C_{\mathrm{series}}} = \frac{t_{\mathrm{f,sl}}}{\varepsilon_\mathrm{f} L} + \frac{y}{\varepsilon_\mathrm{l} L} + \frac{t_{\mathrm{f,st}}}{\varepsilon_\mathrm{f} L}. \tag{6}$$

It is worth noting that the terms in $\boldsymbol{c}_\mathrm{m}$ are in the forms of $\cos^2 \theta_{i,j}$ where $\theta_{i,j}$ is the common angle between the electric field vector and the electrode surface normal, as indicated in Fig. 2. Generally, the capacitance matrix $\mathbf{C}$ contains sinusoidal harmonics, but depending on the assumptions for the electric fields between the charges in the electrodes, higher-order harmonics may be ignored and the matrix is simplified to contain only lower-order harmonics [26], [27]. For small angles, $\theta_{i,j}$ is approximately equal to the actuator position x (assuming $d = 1$), and in this case $\cos^2 \theta_{\mathrm{i,j}} \approx \cos^2 x$ which trigonometrically reduces to $(1 + \cos 2x)/2$, revealing a zeroth and second-order harmonic.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:13:56 UTC from IEEE Xplore. Restrictions apply.

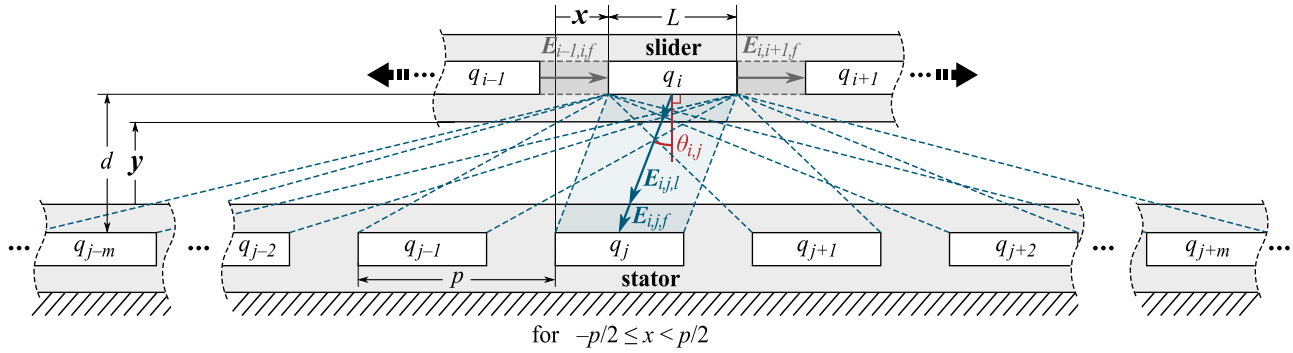


Fig. 2. FPN: The electric fields related to the ith charge in the slider are assumed to be confined to the parallelogram regions that are projections between the m neighboring electrodes of the stator, shown to be bounded by dashed lines. $E_{i,j,f}$ and $E_{i,j,l}$ are the electric fields from q_i to q_j through the insulating film and the separating dielectric lubricant, respectively. $\theta_{i,j}$ is the common angle between the electric field vector $E_{i,j}$ and the electrode surface normal of either q_i or q_j.

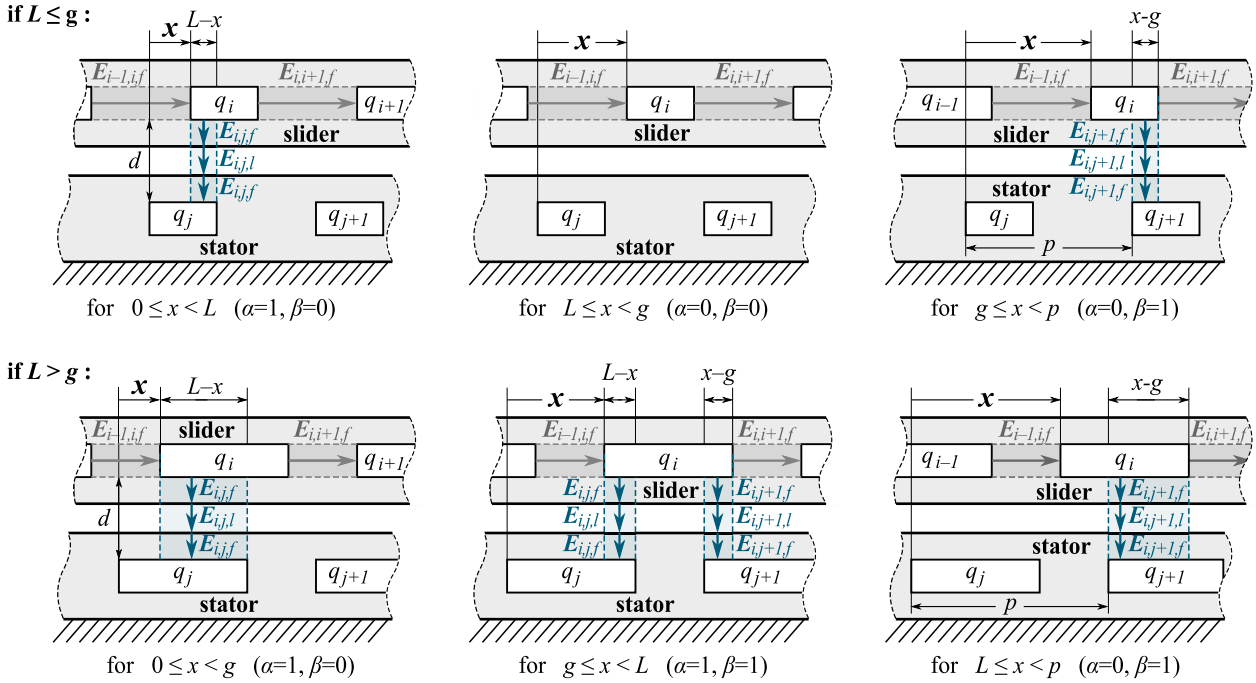


Fig. 3. FSV: Electric fields related to the ith charge in the slider are assumed to be confined to strictly vertical regions projected by electrodes of the stator, shown to be bounded by dashed lines. The piece-wise defined electric fields are shown for $L \le g$ (top) and for $L > g$ (bottom) with the corresponding values for α and β provided below each region. $E_{i,j,f}$ and $E_{i,j,l}$ are the electric fields from q_i to q_j through the insulating film and the separating dielectric lubricant, respectively.

2) Fields Are Strictly Vertical (FSV): Electric fields are confined to strictly vertical regions, resulting in piece-wise defined fields. Only the field from directly overlapping electrodes is considered to be significant. Fig. 3 shows the piece-wise field definitions by the shaded regions bounded by dashed lines. Based on the relative sizes of L and g, the fields are defined separately for different ranges of x and are assigned parameters α and β such that $\alpha = 1$ when the field corresponding to the overlapping area of q_i and q_j exists and $\beta = 1$ when the field corresponding to the overlapping area of q_i and q_{j+1} exists, each otherwise equal to zero. The capacitance terms can then be reduced to

$$\boldsymbol{c}_{\text{s}} = \left(-\frac{\varepsilon_{\text{f}}\, t}{g}, C_{\Sigma}, -\frac{\varepsilon_{\text{f}}\, t}{g} \right) \tag{7}$$

and

$$\boldsymbol{c}_{\text{m}} = C_{\text{series}} \left(0, \alpha \left(1 - \frac{x}{L} \right), \beta \left(\frac{x}{L} - \frac{g}{L} \right) \right) \tag{8}$$

where C_{Σ} is likewise the sum of the absolute values of all the other terms in both $\boldsymbol{c}_{\text{s}}$ and $\boldsymbol{c}_{\text{m}}$ such that

$$C_{\Sigma} = \frac{\varepsilon_{\text{f}}\, t}{g} + \frac{\varepsilon_{\text{f}}\, t}{g} + \alpha \left(1 - \frac{x}{L} \right) + \beta \left(\frac{x}{L} - \frac{g}{L} \right) \tag{9}$$

and C_{series} is the same as (6).

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:13:56 UTC from IEEE Xplore. Restrictions apply.

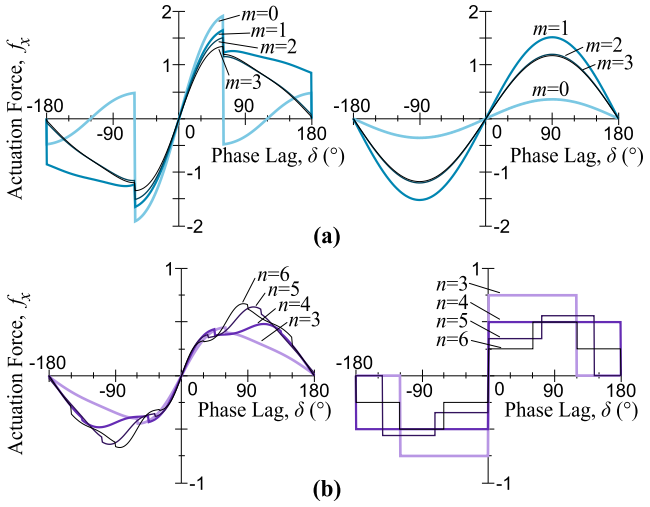


Fig. 4. Cross-sectional actuation force f_x according to phase lag $\delta = \theta_e - \theta_x$ with ε_f, ε_l, V, L, d, p equal to unity. (a) Different neighbor counts m in FPN with electrical angle $\theta_e = 0$ on the left and with forces averaged over all θ_e on the right. (b) Different number of electrical phases n with $\theta_e = 0$ for FPN ($m = 3$) on the left and for FSV on the right. Forces have been normalized by the number of n for comparison.

B. Cross-Sectional Force

In both cases, the gradient of the capacitance matrix $\nabla\mathbf{C}$ becomes

$$\nabla\mathbf{C} = \begin{bmatrix} \nabla C_\Sigma \odot \mathbf{I} & -\nabla\mathbf{C}_m \\ -\nabla\mathbf{C}_m{}^T & \nabla C_\Sigma \odot \mathbf{I} \end{bmatrix} \tag{10}$$

where $\mathbf{I}$ is the identity matrix and $\odot$ is the Hadamard product. After differentiation, the self capacitance matrix $\mathbf{C}_s$ becomes a diagonal matrix, indicating that the force results from mutual capacitance. From (1) and (10), the force for the ith electrode aligned with the jth electrode is then

$$\boldsymbol{f}_{i,j} = \frac{1}{2}\nabla\Big(\boldsymbol{c}_m \cdot ((v_i - v_{j-m})^2, \ldots, (v_i - v_j)^2, \ldots, (v_i - v_{j+m})^2)^T\Big) \tag{11}$$

for FPN when m may be arbitrarily chosen and for FSV when $m = 1$ since only the fields from directly overlapping electrodes are considered. Particularly for FSV, the evaluation of $f_{i,j,x}$ with (8) results in a piece-wise constant relation for constant voltages.

By supplying balanced n-phase voltage waveforms of amplitude V oppositely to the slider and the stator

$$v_i = V \sin\left(\theta_{sl} - \frac{2\pi}{n} i\right) \tag{12}$$

and

$$v_j = -V \sin\left(\theta_{st} - \frac{2\pi}{n} j\right) \tag{13}$$

where θ_{sl} and θ_{st} are the electrical angles of the supplied voltages to the slider and the stator, respectively. Expressing the jth electrode in the stator as the kth offset from the ith electrode in the slider so that $j = i + k$, and summing the force in (11) over one set of phases in the slider (electrodes $i = 1, \ldots, n$; $n \geq 3$) yields

$$\boldsymbol{f}_k = \frac{1}{2} V^2 n \nabla(\boldsymbol{c}_m \cdot \boldsymbol{u}_k) \tag{14}$$

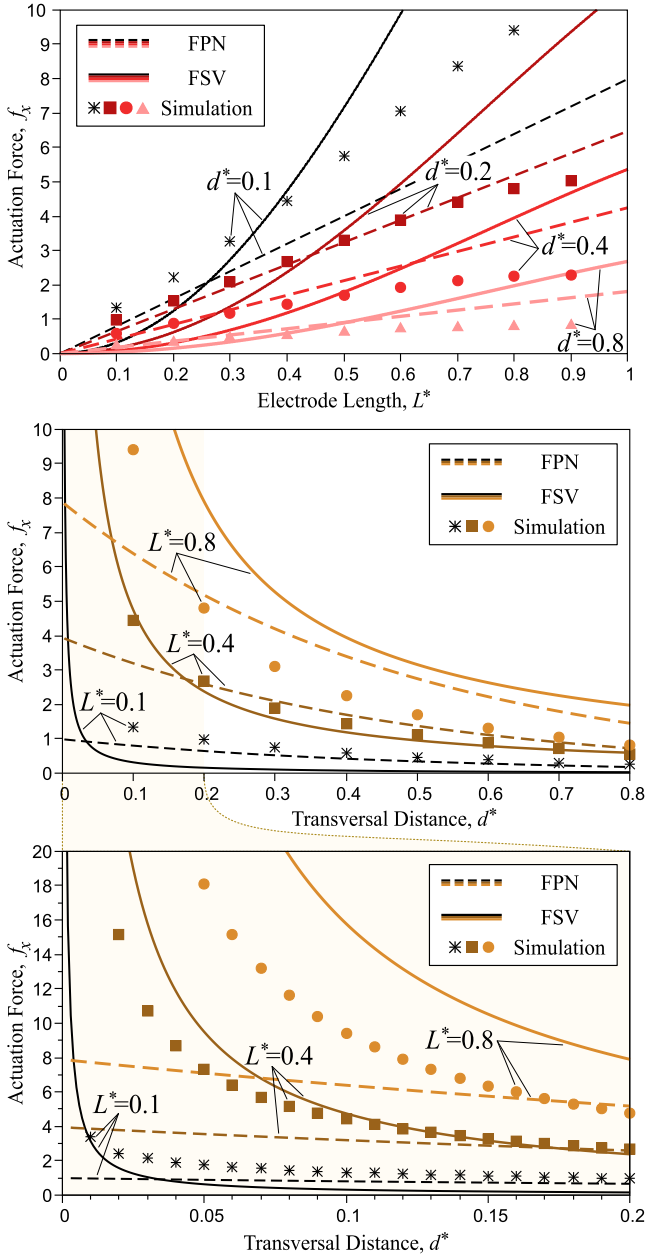


Fig. 5. Comparison of peak force (at $\delta = 90°$) between FPN ($m = 3$, dashed lines), FSV (solid lines), and FEA simulation (points) from COMSOL Multiphysics averaged over all θ_e with ε_f, ε_l, V, p equal to unity. L^* is the electrode length L normalized by pitch p, and d^* is the transversal distance d likewise normalized by the pitch p. For simulation, the electrode thickness was set to a small value ($t = 0.01$). Top: force f_x versus L^* at different d^*. Middle: force f_x versus d^* at different L^*. Bottom: rescaled version of the middle plot with morebr simulation points.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:13:56 UTC from IEEE Xplore. Restrictions apply.

TABLE I
MODEL ERROR

Normalized Electrode Length, L^*	Normalized Transversal Distance, d^*									
	FPN Error					FSV Error				
	0.1	0.2	0.4	0.6	0.8	0.1	0.2	0.4	0.6	0.8
0.1	−0.5245	−0.3285	−0.1651	−0.1056	**−0.0628**	−1.0107	−0.8198	−0.5119	−0.3315	−0.2050
0.2	−0.6273	−0.2232	**−0.0292**	**0.0002**	**−0.0059**	−0.9858	−0.8990	−0.5693	−0.3494	−0.2136
0.4	−1.2286	**−0.0748**	0.2704	0.2327	0.1596	0.3271	−0.2901	−0.2416	**−0.0878**	**0.0284**
0.6	−2.2587	**0.0007**	0.6232	0.5071	0.3527	2.8618	1.0712	0.5520	0.4909	0.5042
0.8	−3.0083	0.4017	1.1501	0.9146	0.6304	6.4052	3.1222	1.7017	1.3240	1.1557

Note: The bold values indicate errors within ±0.1.

where

$$\boldsymbol{u}_k = \begin{pmatrix} 1+\cos(\theta_{\text{st}} - \theta_{\text{sl}} - 2\pi(k-m)/n) \\ \vdots \\ 1+\cos(\theta_{\text{st}} - \theta_{\text{sl}} - 2\pi k/n) \\ \vdots \\ 1+\cos(\theta_{\text{st}} - \theta_{\text{sl}} - 2\pi(k+m)/n) \end{pmatrix} \tag{15}$$

for FPN (m is arbitrary) and FSV ($m = 1$). Because the model is periodic, the offset k is defined as $k = \lfloor x_{\text{pos}}/p + 1/2 \rfloor$ for FPN and $k = \lfloor x_{\text{pos}}/p \rfloor$ for FSV where $\lfloor\ \rfloor$ denotes the floor function. The motor position x_{pos} along the x-axis then converts into the relative position x in the models by using $x = x_{\text{pos}} - kp$.

C. Force With Skewing Patterns

The cross-sectional model is extended along the z direction normal to the plane of the cross-section to provide the full actuator force model. When unskewed, the electrodes extend in a straight manner parallel to the z-axis to a width w. In this case, the motor force $\boldsymbol{F}$ is then simply the product of the cross-sectional force $\boldsymbol{f}$ and the electrode width w.

Generally, however, skewing may be introduced to the electrodes of the slider in the form of a skew pattern function $z = z(\xi)$ where ξ is the coordinate parallel to the x-axis in the frame local to the slider. Because the slider moves with the motor position x_{pos}, the coordinate transformation to the global frame is then $x = x_{\text{pos}} + \xi$, and the skew pattern function expressed in the global x-coordinate becomes $z = z(x - x_{\text{pos}})$. The actuation force F_x is then

$$F_x = \int_{z_0}^{z_{\text{f}}} f_{k,x}\, \mathrm{d}z = \int_{x_0}^{x_{\text{f}}} f_{k,x}(x)\, \zeta(x - x_{\text{pos}})\, \mathrm{d}x \tag{16}$$

where ζ is the derivative of the skew pattern function ($\mathrm{d}z/\mathrm{d}x$). When the skew pattern function is odd ($z(-\xi) = -z(\xi)$), its derivative ζ becomes even ($\zeta(x - x_{\text{pos}}) = \zeta(x_{\text{pos}} - x)$), and the integral becomes a convolution, similar to skewing in magnetic synchronous motors [29].

III. RESULTS

A. Cross-Sectional Actuation Force

Because the electrostatic motor is synchronous in nature (i.e., the electrical speed $\dot{\theta}_{\text{e}}$ matches the mechanical speed $\dot{\theta}_x$) [9], [23], Fig. 4 plots the cross-sectional actuation force f_x versus the motor phase lag δ defined as the difference in angular position of the electrical angle $\theta_{\text{e}} = \theta_{\text{st}} - \theta_{\text{sl}}$ and the motor position $\theta_x = 2\pi x/np$ such that $\delta = \theta_{\text{e}} - \theta_x$.

Fig. 4(a) shows how various m affects the actuation force for FPN. While including an infinite number of neighboring charges ($m \to \infty$) would be ideal for FPN, the increasing computational cost makes such a choice impractical. From comparison, the results for the FPN model when $m = 2$ are similar enough to when $m = 3$, suggesting that the choice of $m = 3$ is sufficient for the scope of this article. Fig. 4(b) shows the actuator force at different numbers of supplied electrical phases n in FPN with $m = 3$ and in FSV. Larger n increases the number of piecewise regions and makes the force curves appear more jittery, but the curves with larger n better resemble the generic sinusoidal force curve of synchronous motors. For the scope of this article, the common three-phase case ($n = 3$) is used for experimentation validation and comparison.

With $m = 3$ and $n = 3$, the peak average forces of FPN and FSV are compared with simulation results (COMSOL Multiphysics) according to the normalized electrode length $L^* = L/p$ and the normalized transversal distance $d^* = d/p$ in Fig. 5. For both FPN and FSV, the actuator force increases with L^* and decreases with d^*. This is consistent with the fact that increasing L^* and decreasing d^* improves the electric field coupling between the slider and the stator electrodes. While FPN provides a linear force with respect to L^*, FSV has a more parabolic relation that starts smaller than FPN but grows and becomes larger at higher L^*. With respect to d^*, both FPN and FSV decrease nonlinearly and rather quickly. Therefore, to achieve large force, large electrode lengths and short transversal distances are desirable. Note, however, that L^* cannot be larger than or equal to unity because in practice that would cause an electrical short circuit. On the other hand, distances d^* larger than unity are possible but provide weak force, and the unilaterial film thicknesses $t_{\text{f,sl}}$, $t_{\text{f,st}}$, and any lubricating dielectric materials put a lower limit on d^*.

Results from the simulation were obtained in 2D with electrode thickness significantly smaller than the electrode length ($L = 1$, $t = 0.01$) to best match the assumptions of the model. The simulation results include fringing effects and therefore differ from the two presented models, but the models still follow the same trends. Table I shows the percent errors of the models in relation to the simulation results. Errors within ±0.1 are indicated in bold text. FPN tends to under-estimate the actuator

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:13:56 UTC from IEEE Xplore. Restrictions apply.

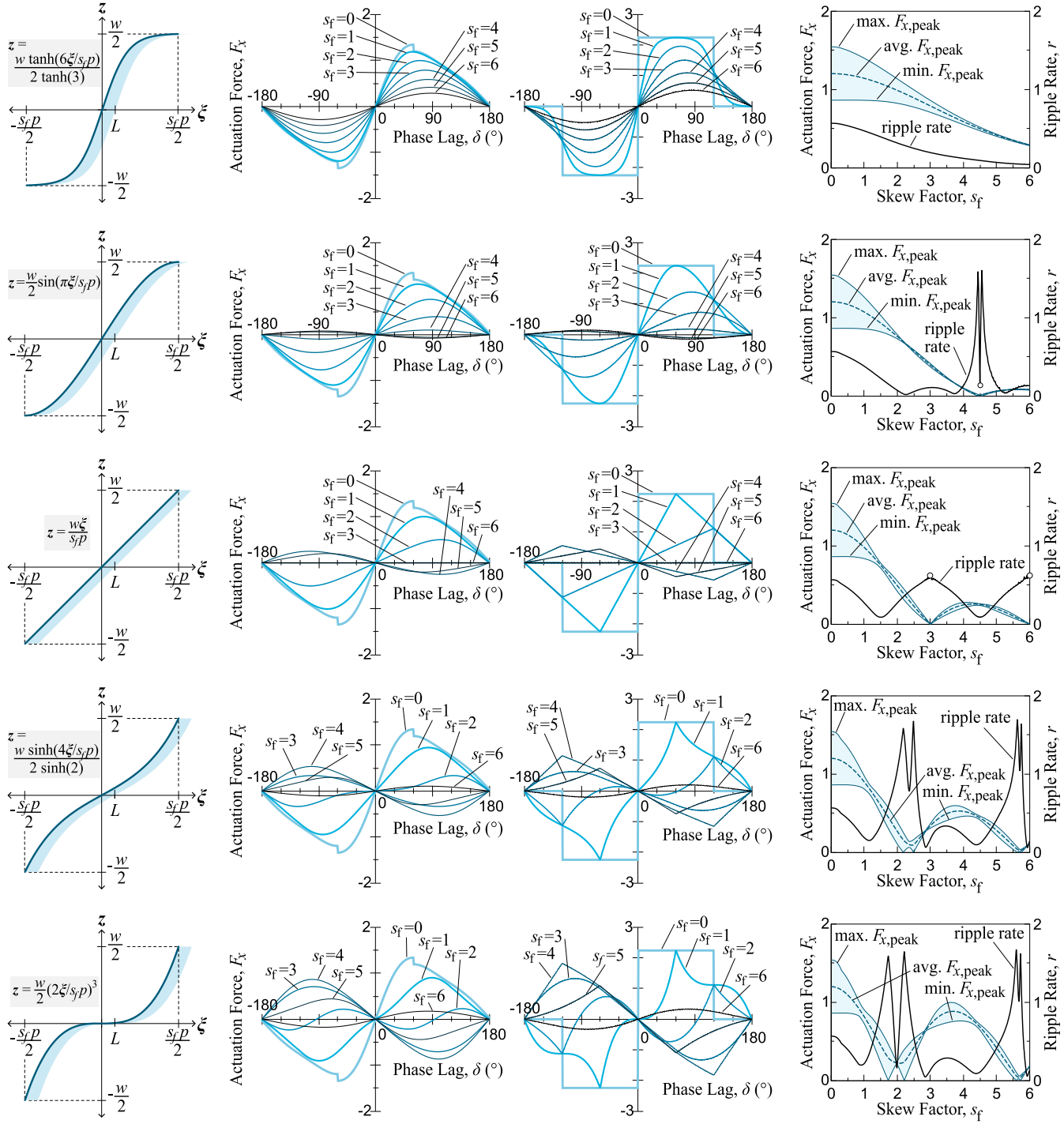


Fig. 6. Left to right: the electrode skew pattern function $z = z(\xi)$; the actuation force F_x for various skew factors s_f at $\theta_e = 0$ for FPN ($m = 3$); the same for FSV; the minimum, average, and maximum peak actuation forces with the force ripple rate at all motor positions x_{pos} for all electrical angles θ_e. Top to bottom: the results for the skew pattern functions $z = w\tanh(6\xi/s_f p)/2\tanh(3)$; $z = w\sin(\pi\xi/s_f p)/2$; $z = w\xi/s_f p$; $z = w\sinh(4\xi/s_f p)/2\sinh(2)$; and $z = w(2\xi/s_f p)^3/2$. Force plots and ripple rates are obtained with ε_f, ε_l, V, L, d, p, w equal to unity and $n = 3$.

force from simulation when the transversal distance d^* is small and tends to over-estimate when d^* is large, agreeing rather closely near $d^* = 0.2$. FSV, however, generally over-estimates the simulated force except for small values of L^* where fringing is more significant. Because the electric fields take on different shapes when the actuator dimensions are changed, the accuracy of the presented models depends on how closely the actual electrical fields resemble the assumed forms. While combining or averaging the force results of the two presented models may be considered with the hopes of improving the model accuracy, doing so would remove the physical implications of assuming the shapes of the electrical fields. Rather, simply assuming field

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:13:56 UTC from IEEE Xplore. Restrictions apply.

shapes that more closely resemble the actual electrical fields would improve accuracy.

B. Actuation Force With Skewing

Defining the skew factor s_f as the ratio of the span of one skew pattern to the electrode pitch, a generic skew pattern function z has a domain $x \in [-s_f p/2, s_f p/2]$ and a range $z \in [-w/2, w/2]$. As representatives, five such skew pattern functions are chosen and presented in the left-most column of Fig. 6. Corresponding plots of the actuator force when $\theta_e = 0$ are shown using FPN in the second column and using FSV in the third column. Notably, because the cross-sectional forces for FSV are piecewise constant as seen from (8), the convolution integral results in a multiplication of the force and the skew pattern function so that the skew pattern shapes are visually observable in the force plots of the third column. For some patterns, large skew factors result in flipped force curves such that a negative phase lag ($\delta < 0$) produces a positive force.

The force ripples in the right-most column of Fig. 6 are measured by the ripple rate r defined as [27]

$$\text{ripple rate, } r = \frac{\text{maximum } F_{x,\text{peak}} - \text{minimum } F_{x,\text{peak}}}{\text{average } F_{x,\text{peak}}} \quad (17)$$

and produce different results as a function of the skew factor s_f for the five skew pattern functions. In all five cases, the average force is largest at $s_f = 0$ (i.e., no skewing) and begins to decrease along with the ripple rate as s_f increases slightly. Therefore, for the skew patterns shown, introducing slight skewing ($s_f < 1$) reduces the force ripple rate at the cost of also reducing the average force.

While for some skew pattern functions large values of s_f result in large average forces, these force curves are flipped and provide negative force for positive phase lag, as mentioned above. Among the patterns, those in the first and the second rows of Fig. 6 are perhaps preferable over the other patterns because their force curves are smoother; however, the force ripple rates for the patterns in the bottom two rows decrease faster with the skew factor s_f than those above and may instead be preferred when small skew factors are desired while maintaining low ripple rates.

IV. Experimental Validation

A. Prototypes and Setup

To validate the model experimentally, three-phase electrostatic actuators were fabricated using FPCB technology. Fig. 7 shows the experimental setup. The electrodes in the stator and the slider were made of copper, and the dielectric filling between the electrodes and the films covering the electrode surfaces were made of polyimide. The actuator parameters, i.e., the overall size, the pitch p, and the normalized electrode length L^*, were chosen based on the dimensions used in previous research [19], [27], [30]. The stator and the slider films were both 80 mm wide and their lengths were 135 and 100 mm, respectively.

A total of 14 different pairs of a stator and a slider were prepared, all with a pitch $p = 300$ μm. Two were patterned

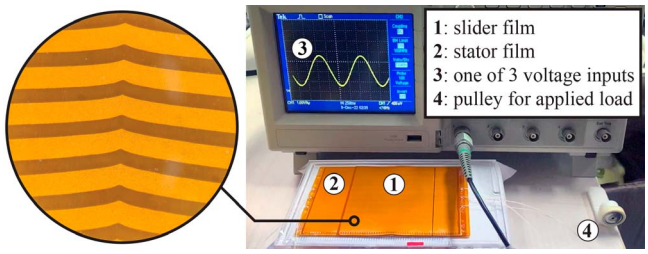


Fig. 7. Experimental setup. The enlarged view of the slider film shows the electrodes with a skew pattern function $z = w \sin(\pi\xi/s_f p)/2$ and a skew factor $s_f = 2.26$.

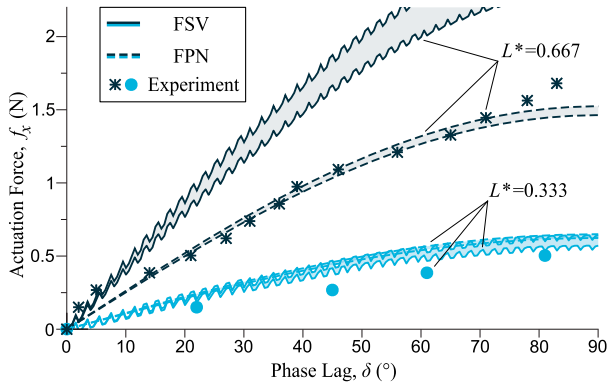


Fig. 8. Experimental data and modeled actuation forces for the straight electrode pattern ($s_f = 0$) using FPN ($m = 3$) and FSV averaged over all θ_e. The shaded regions, colored to match their corresponding experimental data, are the model-predicted force for a dielectric fluid thickness y between 15 and 20 μm and a normalized transversal distance d^* between 0.14 and 0.16.

with straight unskewed electrodes ($s_f = 0$), each with different normalized electrode lengths ($L^* = 0.333$ and 0.667). The rest were designed with a normalized electrode length $L^* = 0.667$. Three used skew pattern $z = w \sin(\pi\xi/s_f p)/2$ in the slider, three used $z = w\xi/s_f p$ in the slider, and three used $z = w \sinh(4\xi/s_f p)/2\sinh(2)$ in the slider, patterns which corresponded to the skew patterns of the middle three rows of Fig. 6. Each skew pattern in the slider had three different skew factors ($s_f = 1$, 2, and 3), while the stator electrodes were kept straight. The last three pairs had sliders with the above three skew patterns but with the skew factors that corresponded to the first minimum ripple rate of each pattern (Fig. 6). The skew patterns were generated as splines using a function plotter plugin inside an open source vector program (Inkscape) and then imported into an open source PCB design program (KiCAD) where it was patterned and routed before being sent for fabrication.

Every set of three electrodes in each stator or slider was electrically connected to each other to accommodate three-phase voltage input. To drive the actuator, the electrodes of the stator were connected in the opposite order to the electrodes of the slider, using the wiring method first introduced in [30] to reduce the number of high-voltage inputs required by half. For our prototypes, this resulted in reducing the required six inputs

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:13:56 UTC from IEEE Xplore. Restrictions apply.

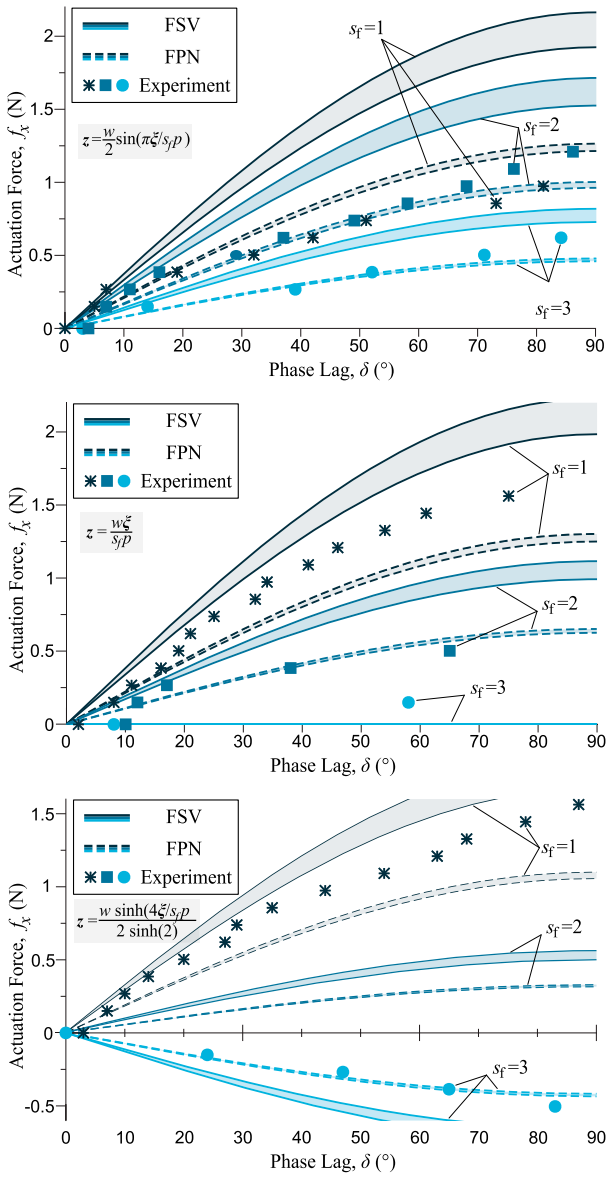


Fig. 9. Experimental data and modeled actuation forces for skewed electrode patterns using FPN ($m = 3$) and FSV averaged over all θ_e. Top to bottom: the results for the electrodes with three skew pattern functions, $z = w\sin(\pi\xi/s_f p)/2$, $z = w\xi/s_f p$, and $z = w\sinh(4\xi/s_f p)/2\sinh(2)$. The shaded regions, colored to match their corresponding experimental data, are the model-predicted force for a dielectric fluid thickness y between 15 and 20 μm and a normalized transversal distance d^* between 0.14 and 0.16.

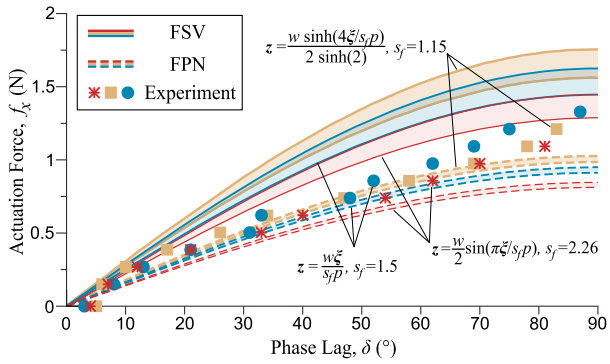


Fig. 10. Experimental data and modeled actuation forces for three skew patterns, $z = w\sin(\pi\xi/s_f p)/2$, $z = w\xi/s_f p$, and $z = w\sinh(4\xi/s_f p)/2\sinh(2)$, at skew factors that minimize the ripple rates. Both FPN ($m = 3$) and FSV were averaged over all θ_e. The shaded regions, colored to match their corresponding experimental data, are the model-predicted force for a dielectric fluid thickness y between 15 and 20 μm and a normalized transversal distance d^* between 0.14 and 0.16.

to three. The stator and slider pair was placed in a flat container with a dielectric liquid applied between the films, after which they were driven by three-phase alternating high voltage inputs at a frequency of 1 Hz and a zero-to-peak magnitude of 1200 V.

The motor prototypes were experimentally characterized in the region for stable motor operation ($\delta \in [0°, 90°]$) by driving the synchronous actuator under incrementally different constant loads. A paper cup was attached to one end of the slider through a string and suspended against gravity by a pulley. Weights were added to the cup such that the entire load incremented by 0.118 N. The actuator was driven in the direction that would lift the load, and the relative positions of the slider to the stator were recorded using a high-speed camera while the input voltages were monitored through an oscilloscope (TDS2014C, Tektronix). Afterward, the two were converted to the motor position θ_x and the electrical angle θ_e from which the phase lag δ was calculated. Due to the opposite wiring method mentioned above, $\theta_{sl} = -\theta_{st}$ and so the electrical angle θ_e became twice the measured angle.

B. Comparison and Validation

Fig. 8 shows the actuator prototype forces for the straight, unskewed electrode patterns with two different normalized electrode lengths $L^* = 0.333$ and $L^* = 0.667$. We used a reasonable range of the dielectric thickness to calculate the model-predicted force because the fluid thickness could not be precisely measured in the experiments. From comparison, the model using FPN better estimated the results for $L^* = 0.667$ while the model using FSV over-estimated the results. Both models well estimated the results for $L^* = 0.333$. This is consistent with the comparisons in Fig. 5, where FSV estimated larger force at larger L^* than FPN.

Fig. 9 compares the results of three of the skew patterns with our model using FPN and FSV. Overall, the experimental results fall within the model's estimations for both FPN and FSV, although the model using FSV is slightly higher as $L^* = 0.667$ is used. It is also worth noting that the three skewed prototypes produce distinctly different actuation forces as the skew factor s_f changes, highlighting the importance of being able to model the effects of skew before production.

Fig. 10 shows the experimental and the model results for the three skew patterns with their skew factors each minimizing their respective ripple rates. With these optimized skew factors, both experimental and model results agree that the actuation forces do not differ greatly from those of the other patterns. In this case, there is some freedom in choosing a skew pattern

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:13:56 UTC from IEEE Xplore. Restrictions apply.

that is easier to fabricate while keeping the same overall performance, as long as the corresponding skew factor is used.

V. Conclusion

We presented a model for linear electrostatic motors with repeatedly patterned electrodes in a slider and stator configuration. Importantly, the model provides the physical derivation of the capacitance matrix and the motor force from mutual capacitance, the accuracy of which depends on how closely the assumed electrical fields resemble the actual electrical fields formed by the actuator with the given design parameters. Cross-sectional actuation force is explored by changing the neighbor count m, the number of phases n, and by comparing the peak average force for different normalized electrode lengths L^* and normalized transversal distances d^*. Larger L^* and smaller d^* give higher forces, but both have their limits when put in practice. Comparison of the two models with simulation results reveals that the assumed shapes of the electrical fields are relatively accurate for different ranges of the actuator design parameters. Extending the model to the width-dimension allows the inclusion of generic skewing patterns, five of which were shown for their actuation force and force ripple rates at different skew factors. FPN and FSV were compared with the experimental results of straight electrodes of different lengths, showing that FPN had better force estimations for both small and large L^* with the given d^*. The experimental results of three of the skew patterns were within the model's predictions, supporting its validity. An estimated range was used for the dielectric fluid thickness because it was not precisely measurable, but future work with proper instrumentation could better improve the evaluation of the model.

Additionally, although the results provided here were normalized by many parameters, such as ε_f, ε_l, V, L, d, p, and w, the model does not lose generality when used with other more practical values and dimensions in actual linear electrostatic motors and prototypes. Design parameters beyond the specific examples in this article are fully acceptable in the model, and therefore the work presented in this article contributes to the development of more optimal and readily applicable electrostatic motors for devices and robotics. Furthermore, the modeling approach presented here is applicable to any generic skew pattern that may be analytically written. Since skewing is a design technique that is used in other patterned actuators, such as traditional rotary motors, we hope the work here acts as a reference for future patterned actuator design models to follow and expand upon.

References

[1] R. Borgani, D. Forchheimer, J. Bergqvist, P.-A. Thorén, O. Inganäs, and D. B. Haviland, "Intermodulation electrostatic force microscopy for imaging surface photo-voltage," *Appl. Phys. Lett.*, vol. 105, no. 14, 2014, Art. no. 143113.

[2] Q. Zhou and A. Zettl, "Electrostatic graphene loudspeaker," *Appl. Phys. Lett.*, vol. 102, no. 22, 2013, Art. no. 223109.

[3] G. Gramse, M. Edwards, L. Fumagalli, and G. Gomila, "Dynamic electrostatic force microscopy in liquid media," *Appl. Phys. Lett.*, vol. 101, no. 21, 2012, Art. no. 213108.

[4] K.-i. Umeda, K. Kobayashi, K. Matsushige, and H. Yamada, "Direct actuation of cantilever in aqueous solutions by electrostatic force using high-frequency electric fields," *Appl. Phys. Lett.*, vol. 101, no. 12, 2012, Art. no. 123112.

[5] Y. Ma, Y.-Z. Bai, H.-Y. Li, Z.-B. Zhou, and Z. Zhou, "Identification and compensation of quadratic terms of a space electrostatic accelerometer," *Rev. Sci. Instrum.*, vol. 89, no. 11, 2018, Art. no. 114502.

[6] O. Ozkan, A. Shahriari, and V. Bahadur, "Electrostatic suppression of the Leidenfrost state using AC electric fields," *Appl. Phys. Lett.*, vol. 111, no. 14, 2017, Art. no. 141608.

[7] T. Liu, S. Wang, F. Han, and Q. Wu, "Modeling and compensation of cross-axis coupling in an electrostatic accelerometer for testing the equivalence principle," *Rev. Sci. Instrum.*, vol. 89, no. 12, Art. no. 124501, 2018.

[8] S. Tapuchi and D. Baimel, "Novel multilayer differential linear electrostatic motor," in *Proc. Int. Symp. Power Electron., Electr. Drives, Autom. Motion*, 2014, pp. 1368–1372.

[9] M. Dadkhah, Y. Hojjat, J. Jeon, M. Ghodsi, and M. Modabberifar, "Voltage-induction synchronous electrostatic motor," *Int. J. Adv. Manuf. Technol.*, vol. 77, no. 1, pp. 145–164, 2015.

[10] T. Liu, S. Wang, F. Han, and Q. Wu, "Rotation control of a variable-capacitance electrostatic motor for space equivalence principle tests with rotating masses," *Rev. Sci. Instrum.*, vol. 91, no. 1, 2020, Art. no. 015003.

[11] Z. G. Zhang, M. Gondo, N. Yamashita, A. Yamamoto, and T. Higuchi, "Design and control of a fish-like robot using an electrostatic motor," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2007, pp. 974–979.

[12] H. Wang, A. Yamamoto, and T. Higuchi, "Electrostatic-motor-driven electroadhesive robot," in *Proc. IEEE/RSJ Int. Conf. Intell. Robot. Syst. (IROS)*, 2012, pp. 914–919.

[13] H. Wang and A. Yamamoto, "Analyses and solutions for the buckling of thin and flexible electrostatic inchworm climbing robots," *IEEE Trans. Robot.*, vol. 33, no. 4, pp. 889–900, Aug. 2017.

[14] H. Wang and A. Yamamoto, "A thin electroadhesive inchworm climbing robot driven by an electrostatic film actuator for inspection in a narrow gap," in *Proc. IEEE Int. Symp. Saf. Secur. Rescue Robot. (SSRR)*, 2013, pp. 1–6.

[15] Y. Kim and H.-S. Park, "The switchable cable-driven mechanism to control multiple cables individually using a single motor," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 4376–4383, Apr. 2022.

[16] R. Hinchet and H. Shea, "High force density textile electrostatic clutch," *Adv. Mater. Technol.*, vol. 5, no. 4, 2020, Art. no. 1900895.

[17] R. Hinchet and H. Shea, "Glove- and sleeve-format variable-friction electrostatic clutches for kinesthetic haptics," *Adv. Intell. Syst.*, vol. 4, no. 12, 2022, Art. no. 2200174.

[18] Y. Kim, J. Kim, D. Lee, J. Piao, J. Bae, and S. H. Koo, "Development of clothing-type platforms considering pressure and user satisfaction: Focusing on industrial workers who tend to lift loads," *Textile. Res. J.*, vol. 93, no. 9–10, pp. 2226–2241, 2023.

[19] I. Choi, S. J. Yoon, and Y.-L. Park, "Linear electrostatic actuators with Moiré-effect optical proprioceptive sensing and electroadhesive braking," *Int. J. Robot. Res.*, vol. 43, no. 5, pp. 646–664, 2023.

[20] F. Kimura, A. Yamamoto, and T. Higuchi, "FPGA implementation of a signal synthesizer for driving a high-power electrostatic motor," in *Proc. IEEE Int. Symp. Ind. Electron.*, 2011, pp. 1295–1300.

[21] X. Li and A. Yamamoto, "A multi-slider linear actuator using modulated electrostatic attraction and inertia effect," in *Proc. 11th France-Jpn. 9th Europe-Asia Congr. Mechatron. (MECATRONICS)/17th Int. Conf. Res., Educ. Mechatron. (REM)*, 2016, pp. 296–297.

[22] M. Osada, G. Zhang, F. Carneiro, M. Muromachi, S. Yoshimoto, and A. Yamamoto, "Effect of insulating liquid viscosity on output force of a high-power synchronous electrostatic motor," *Int. J. Mech. Sci.*, vol. 220, Apr. 2022, Art. no. 107149.

[23] T. Hosobata, A. Yamamoto, and T. Higuchi, "Transparent synchronous electrostatic actuator for long-stroke planar motion," *IEEE/ASME Trans. Mechatron.*, vol. 20, no. 4, pp. 1765–1776, Aug. 2014.

[24] A. Yamamoto, T. Niino, T. Ban, and T. Higuchi, "A high-power electrostatic motor using skewed electrodes," *Electr. Eng. Jpn.*, vol. 125, no. 3, pp. 50–58, 1998.

[25] F. Kimura, M. Gondo, N. Yamashita, A. Yamamoto, and T. Higuchi, "Capacitive-type flexible linear encoder with untethered slider using electrostatic induction," *IEEE Sens. J.*, vol. 10, no. 5, pp. 972–978, May 2010.

[26] T. Hosobata, A. Yamamoto, and T. Higuchi, "Modeling and analysis of a linear resonant electrostatic induction motor considering capacitance imbalance," *IEEE Trans. Ind. Electron.*, vol. 61, no. 7, pp. 3439–3447, Jul. 2014.

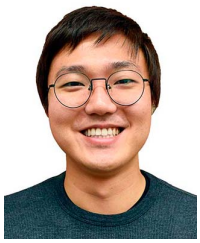
Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:13:56 UTC from IEEE Xplore. Restrictions apply.

[27] A. Yamamoto, T. Niino, and T. Higuchi, "Modeling and identification of an electrostatic motor," *Precis. Eng.*, vol. 30, no. 1, pp. 104–113, 2006.
[28] W. Wang, D. Fan, R. Zhu, P. Wang, Y. Zhao, and H. Wang, "Modeling and optimization of electrostatic film actuators based on the method of moments," *Soft Robot.*, vol. 8, no. 6, pp. 651–661, 2021.
[29] V. Simon-Sempere, M. Burgos-Payan, and J.-R. Cerquides-Bueno, "Spatial filtering: A tool for selective harmonics elimination in the design of permanent-magnet synchronous motors," *IEEE Trans. Magn.*, vol. 48, no. 6, pp. 2056–2067, Jun. 2012.
[30] T. Niino, S. Egawa, and T. Higuchi, "High-power and high-efficiency electrostatic actuator," in *Proc. IEEE Int. Conf. Micro Electro Mech. Syst. (MEMS)*, 1993, pp. 236–241.



Sohee John Yoon received the B.S. degree in mechanical engineering in 2017 from Seoul National University, Seoul, Korea, where he is working toward the Ph.D. degree in mechanical engineering.

His research interests include soft structures, soft sensors, and soft actuators.



Inrak Choi received the Ph.D. degree in mechanical engineering from Stanford University, CA, USA, in 2019.

He was a Postdoctoral Researcher with Soft Robotics Research Center, Seoul National University, Seoul, Korea, while serving the Military in South Korea, from 2019 to 2022. He is currently a robotic algorithms and controls engineer with Intuitive Surgical, CA, USA. His research interests include novel actuation and mechanisms for virtual reality and physical human-robot interaction.



Yong-Lae Park received the Ph.D. degree in mechanical engineering from Stanford University, Stanford, CA, USA, in 2010.

He is currently a Professor with the Department of Mechanical Engineering, Seoul National University, Seoul, Korea. He was an Assistant Professor with the Robotics Institute, Carnegie Mellon University, Pittsburgh, PA, USA, from 2013 to 2017 and a Technology Development Fellow with Wyss Institute for Biologically Inspired Engineering, Harvard University, Cambridge, MA, USA, from 2010 to 2013. His research interests include soft robots, artificial skin sensors and muscle actuators, and soft wearable robots and devices.

Authorized licensed use limited to: Southern University of Science and Technology. Downloaded on July 12,2026 at 13:13:56 UTC from IEEE Xplore. Restrictions apply.